

* Oefeningen reeks 7

1.

* Non-generating symbolen elimineren

- generating symbols = $\{a, b, S, A, C\}$
- non-generating symbols = $\{B\}$

Dus:

$$S \rightarrow CA$$

$$A \rightarrow a$$

$$C \rightarrow b$$

* Non-reachable symbolen elimineren

- Reachable symbols = $\{S, C, A, a, b\}$

Dus de equivalente grammatica zonder nutteloze symbolen is:

$$P = \begin{aligned} S &\rightarrow CA \\ A &\rightarrow a \\ C &\rightarrow b \end{aligned}$$

$$G = (\{S, A, C\}, \{a, b\}, P, S)$$

2.

$$a) n(G) = \{S\}$$

$$S \rightarrow ASB \mid AB$$

$$A \rightarrow aAS \mid aA \mid a$$

$$B \rightarrow SbS \mid bS \mid Sb \mid b \mid A \mid bb$$

b) Unit Pairs =

$$\{(S, S), (A, A), (B, B), (B, A)\}$$

• Unit Producties elimineren

$$S \rightarrow ASB \mid AB$$

$$A \rightarrow aAS \mid aA \mid a$$

$$B \rightarrow SbS \mid bS \mid Sb \mid b \mid bb \mid aAS \mid aA \mid a$$

c) Nee, er zijn geen nutteloze symbolen.

$$d) P \rightarrow a$$

$$Q \rightarrow b$$

$$S \rightarrow AC_1 \mid AB$$

$$C \rightarrow SB_1$$

$$A^1 \rightarrow PC_2 \mid PA \mid a$$

$$C_2 \rightarrow AS$$

$$B^2 \rightarrow SC_3 \mid QS \mid SQ \mid b \mid QQ \mid PC_2 \mid PA \mid a$$

$$C_3 \rightarrow QS$$

3.

S			
-	B		
S	-	C, B	
A	B	B	A, C
a	b	b	c

ab, b

$S \times B$

a, bb

$A \times -$

bb, c

$- \times A, C$

b, bc

$B \times C, B$

$= BC, BB$

a, bbc

$A \times B$

ab, bc

$S \times C, B$

abb, c

$- \times A, C$